Pixel-wise Terrain Segmentation of Mars Rover Images with U-Net

A reproducible control-versus-augmentation study on AI4Mars

Project report | 27 October 2026 | NVIDIA GeForce RTX 4060 Laptop GPU

Summary

This project trains a compact U-Net from scratch to segment Mars rover imagery into soil, bedrock, sand, and big rock. A fixed 80/20 split of the labeled Hugging Face training partition supports a matched comparison between a no-augmentation control and a light paired-augmentation run. The majority-class baseline predicts bedrock everywhere. Validation mIoU selects the final checkpoint; the held-out strict test partition is evaluated once after selection.

Validation selected the no augmentation model at epoch 20 (mIoU 0.771). On the one-shot test_min3 evaluation, the selected model reached mIoU 0.815 and pixel accuracy 0.970; the same-split majority baseline reached mIoU 0.056 and pixel accuracy 0.224.

Model / predictor	Validation mIoU	Best epoch	Logged epoch time
Majority baseline	0.123	-	-
Control	0.771	20	21.4 min
Augmentation	0.761	19	30.5 min

Research question

Under a fixed split and training setup, does light, label-safe augmentation improve the validation performance of a compact U-Net over a control trained without added augmentation? The conclusion is intentionally limited to this dataset version, split, and single seed.

1. Data and label preparation

The project uses the public `hassanjbara/AI4MARS` Hugging Face dataset, a packaged subset of the AI4Mars rover-image collection [1, 2]. The dataset card states a CC0-1.0 license. The source mask IDs are explicitly mapped to four terrain classes. Unknown and void pixels are assigned ignore index 255 and excluded from both loss and metrics.

Partition	Records	Use
Hugging Face train source	18,130	Candidate records
Usable labeled records	15,901	After image/mask checks
Training	12,720	Model fitting and majority class
Validation	3,181	Checkpoint and augmentation selection
Official test_min3	322	One final evaluation after selection

Class IDs and training pixel distribution

ID	Terrain class	Training pixels	Share
0	Soil	97,373,901	37.1%
1	Bedrock	131,047,428	49.9%
2	Sand	31,685,933	12.1%
3	Big rock	2,430,590	0.9%

The majority predictor is **Bedrock** (class 1). On validation it scores pixel accuracy 0.494 but mIoU 0.123; the gap illustrates why overall accuracy alone is misleading under class imbalance.

Split caveat: the packaged schema exposes no scene or rover-sequence grouping key. The saved split is a deterministic record-level split (seed 0), so near-duplicate imagery across partitions cannot be ruled out. The three official test_min variants are alternative annotation-agreement versions of the same test images; only strict test_min3 is used for final scoring [2].

2. Model and experimental protocol

A compact U-Net is trained from random initialization. Repeated 3x3 convolution, batch normalization, and ReLU blocks encode image features; max pooling reduces resolution, while transposed convolutions decode class logits. Skip connections join encoder and decoder features to restore spatial detail, following the original U-Net design [3].



Figure 1. Simplified encoder-decoder layout; widths are feature channels.

Setting	Value
Input / output	192x192 grayscale image / four pixel classes
Base width	32 channels; three downsampling stages
Loss / optimizer	Cross-entropy with ignore_index=255 / Adam
Learning rate / seed	0.001 / 0
Batch / workers / epochs	16 / 2 / 20
Checkpoint rule	Save best validation mIoU checkpoint
Control	Resize and normalize only
Augmentation	Paired horizontal flip and rotation within -8 to +8 degrees; image-only brightness/contrast 0.9-1.1

The control and augmentation runs share the same seed, saved train/validation indices, image resolution, U-Net width, optimizer, batch size, worker count, epoch cap, loss, and best-checkpoint rule. Geometric transforms are applied identically to image and mask; masks use nearest-neighbor interpolation and ignore-index fill. Validation mIoU is the sole model-selection metric; test_min3 is not used for tuning.

The 192x192 input size was selected for local speed after a 256x256 pilot. The completed control run on the RTX 4060 used about 21.4 minutes of logged epoch time for 20 epochs. The augmentation pipeline adds CPU-side image transforms and takes longer per epoch; its cost is reported alongside its validation benefit.

3. Validation comparison

The figures and tables below use only the saved validation partition. Each reported checkpoint is the best checkpoint from its run, selected by validation mIoU.

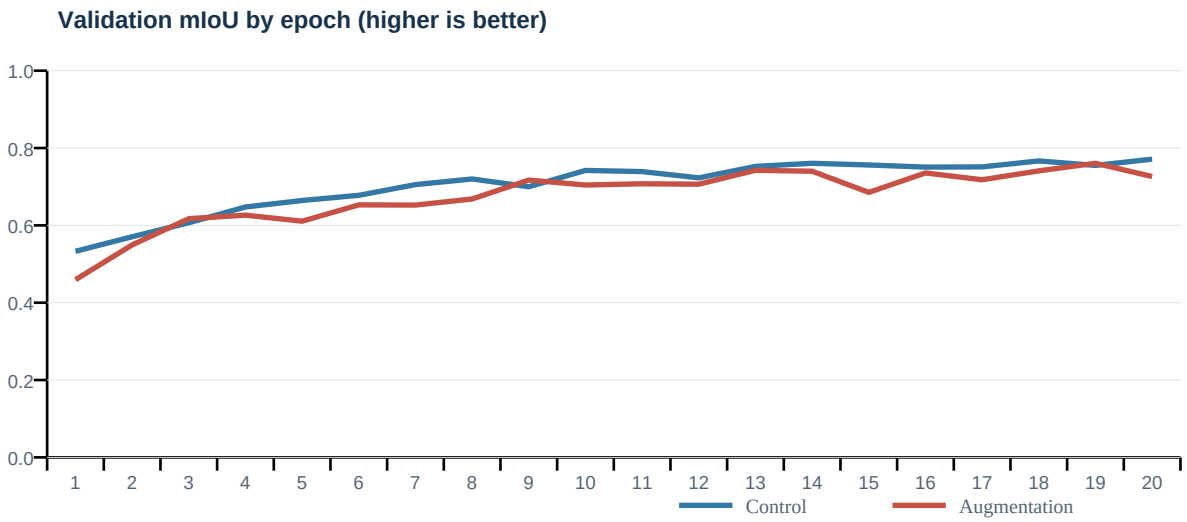


Figure 2. Validation mIoU across the matched 20-epoch runs.

Run	Best epoch	mIoU	Pixel accuracy	Epoch time
Control	20	0.771	0.955	21.4 min
Augmentation	19	0.761	0.944	30.5 min
Majority	-	0.123	0.494	-

Per-class validation IoU

Class	Control	Augmentation
Soil	0.925	0.902
Bedrock	0.931	0.914
Sand	0.854	0.845
Big rock	0.376	0.381

Validation selected **no augmentation** at epoch 20: mIoU 0.771, pixel accuracy 0.955. The majority baseline mIoU is 0.123. The comparison supports a narrow conclusion about this single fixed setup, not a general claim that augmentation always helps or hurts.

4. Qualitative validation examples

Eight fixed validation records are shown with the same input and reference mask beside each model prediction. These examples are drawn from validation only. Overlays use brown for soil, gray for bedrock, tan for sand, red for big rock; void pixels remain uncolored.

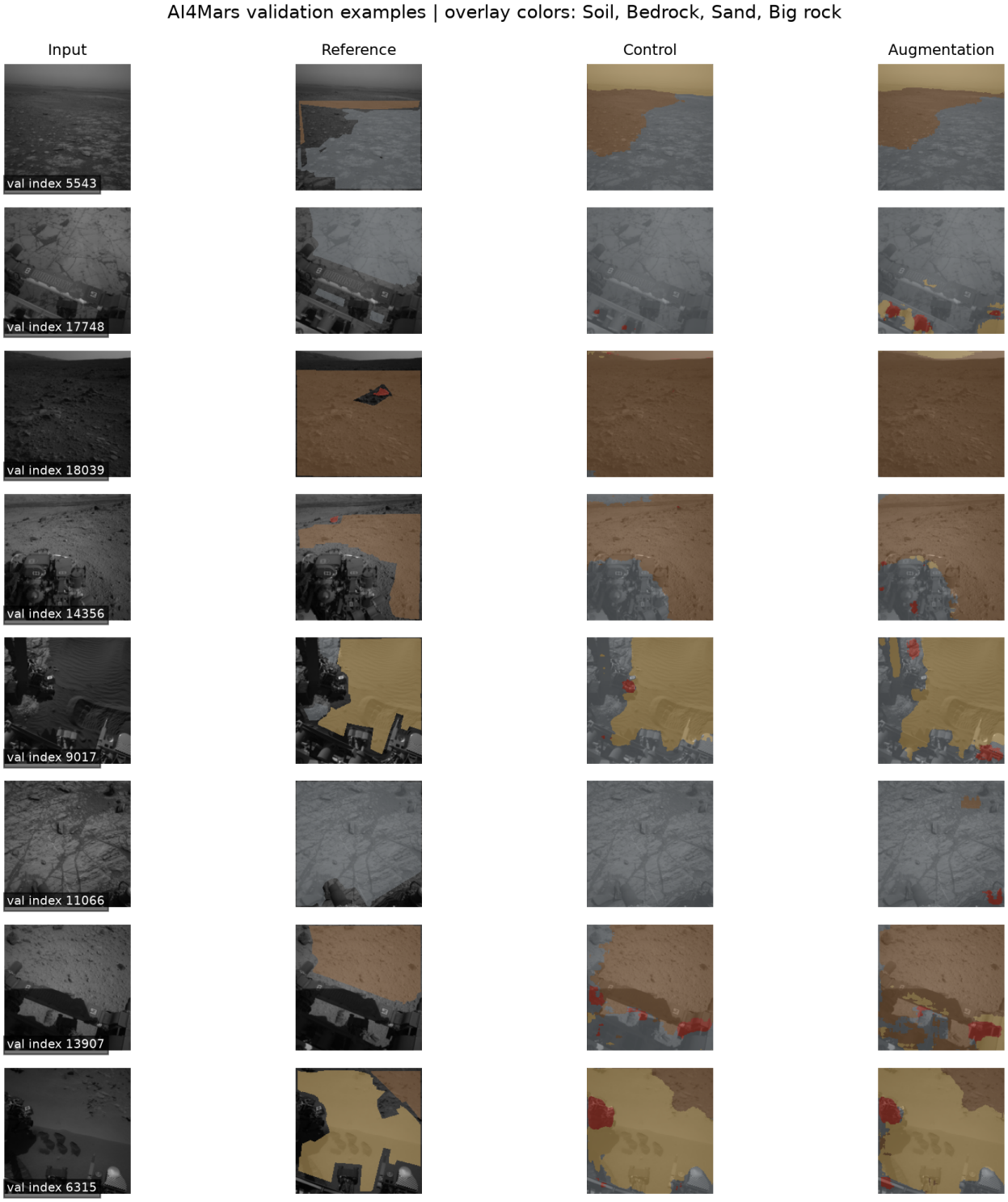


Figure 3. Input, reference, control prediction, and augmentation prediction for eight validation images. The paired comparison helps reveal boundary errors and missed small rocks.

Qualitative samples are illustrative rather than a substitute for aggregate classwise IoU. The validation analysis also records a fixed success and difficult example for the runnable Colab walkthrough.

5. One-shot held-out evaluation

After selecting no augmentation using validation alone, the frozen checkpoint was evaluated once on test_min3 (322 records; 4,112,072 valid non-void pixels). Every record had an image and mask. The majority class was calculated from training pixels only and scored on these same resized test masks in the same evaluation pass.

Predictor	mIoU	Pixel accuracy	Valid pixels
Selected U-Net	0.815	0.970	4,112,072
Majority (Bedrock)	0.056	0.224	4,112,072

Per-class test IoU and confusion matrix

Class	IoU
Soil	0.972
Bedrock	0.901
Sand	0.925
Big rock	0.464

True / Predicted	Soil	Bedrock	Sand	Big rock
Soil	2,085,044	8,327	18,537	78
Bedrock	28,941	889,409	396	2,157
Sand	4,659	58,046	1,013,104	664
Big rock	108	0	0	2,602

Limitations

This is one seed and one train/validation split; the source schema lacks a group key, so related rover scenes may cross the record-level split. The 192x192 input loses fine detail, and the rare big-rock class has far fewer pixels than bedrock. Test_min3 is an expert-agreement subset and may not represent all crowdsourced labels or mission environments. The model is a course-scale experiment, not a validated rover-driving system.

References

[1] R. Michael Swan et al. 'AI4MARS: A Dataset for Terrain-Aware Autonomous Driving on Mars.' CVPR Workshops, 2021. Paper.

[2] hassanjbara/AI4MARS, Hugging Face dataset card (schema, masks, and test annotation variants). Dataset card.

[3] O. Ronneberger, P. Fischer, and T. Brox. 'U-Net: Convolutional Networks for Biomedical Image Segmentation.' MICCAI, 2015. arXiv:1505.04597.

6. Held-out test overlays

These eight examples were captured as part of the single final test evaluation after model selection. They are for post-selection error analysis only and were not used to revise the model.

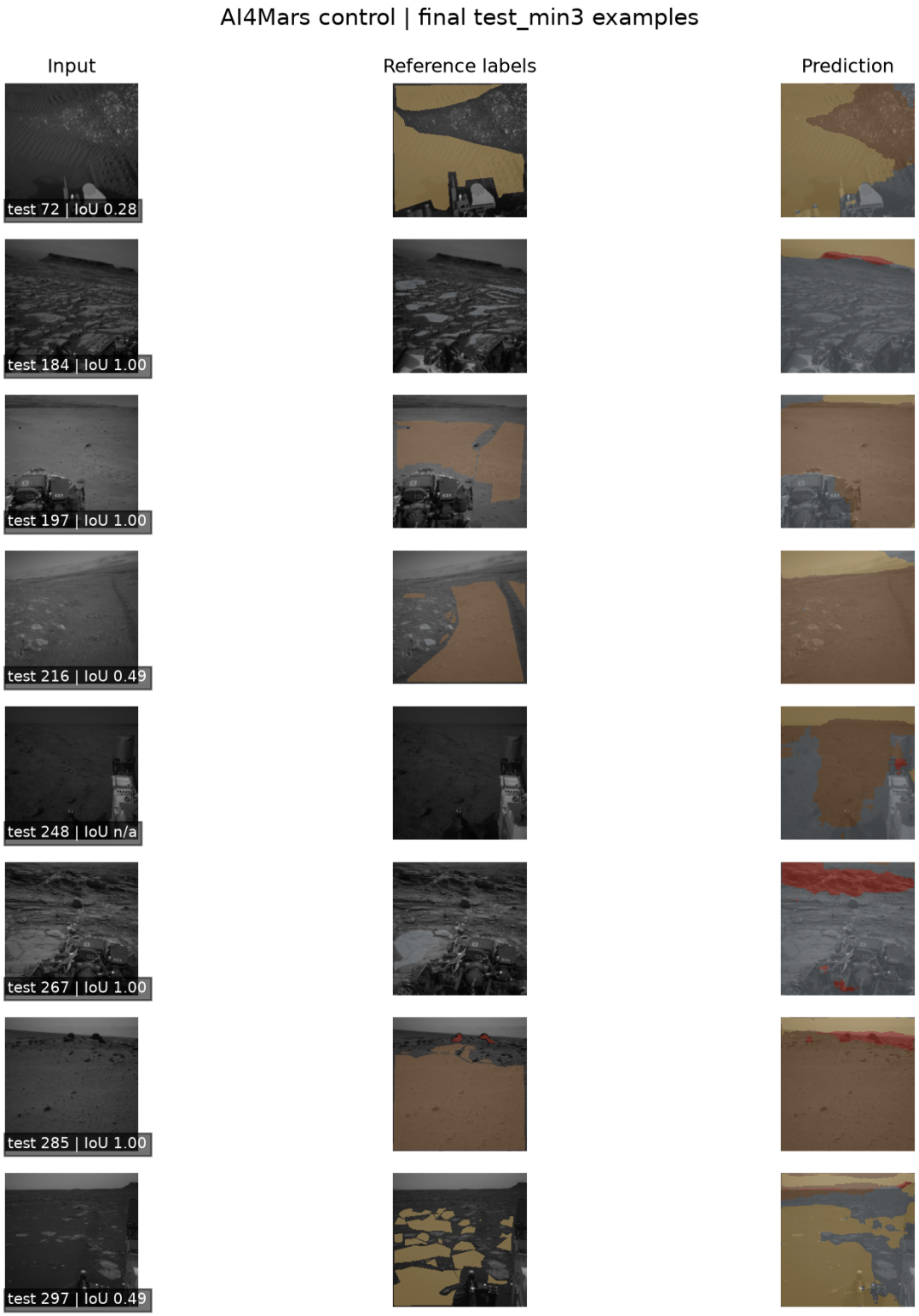


Figure 4. Final selected model predictions on test_min3. Rows were chosen with a fixed random seed; the full aggregate test metrics are in the table on page 6. One displayed strict-annotation record has no valid pixels, shown as IoU n/a; it contributes no pixels to aggregate metrics.